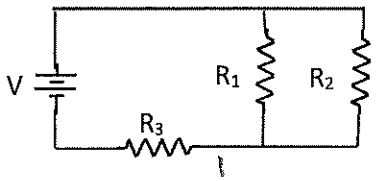
Worksheet # 4 Part B Upload solutions
Physics II

Name: Key Date: _____

Show all work, no work no credit

Upload the solution to questions 11 through 15.

11) In the diagram below, determine the current through R_2?



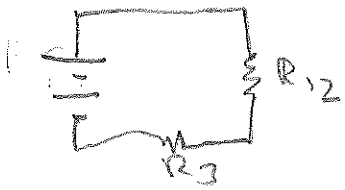
V = 24 V
R_1 = 300 Ω
R_2 = 200 Ω
R_3 = 500 Ω

(1) $R_{12} = \frac{1}{\left(\frac{1}{R_1} + \frac{1}{R_2}\right)} = 120\Omega$

$R_{123} = R_{12} + R_3 = 620\Omega$

$I_{supply} = \frac{V}{R_{123}} = 0.039A$

(2) Using R_{12} to calculate voltage across R_1 and R_2

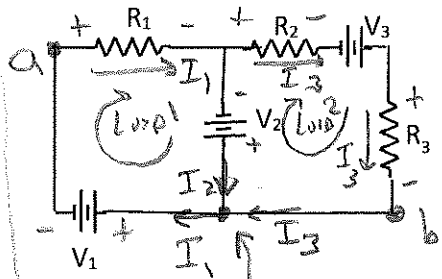


$V_{12} = I_{supply} R_{12} = 4.68V$

$I_2 = \frac{V_{12}}{R_2} = 0.023A$

12) Use Kirchhoff's rules, solve for the current through each of the resistors R_1, R_2 and V_2.

$I_1 = I_2 + I_3$ → loop rule ←



V_1 = 200 V
V_2 = 300 V
V_3 = 150 V
R_1 = 500 Ω
R_2 = 300 Ω
R_3 = 500 Ω

Loop #1 - start at "a"

$(-I_1 R_1) + (+V_2) + (-V_1) = 0$

Solve for I_1

$I_1 = \frac{V_2 - V_1}{R_1} = 0.2A$

$I_1 = I_2 + I_3$

Loop #2 - start at "b"

$(-V_2) + (-I_3 R_2) + V_3 + (-I_3 R_3) = 0$

Solve for I_3

$I_3 (R_2 + R_3) = V_3 - V_2$

$I_3 = \frac{V_3 - V_2}{R_2 - R_3} = -0.1875A$ (picked wrong direction)

$I_1 = I_2 + I_3$

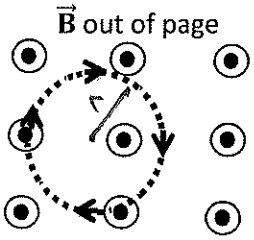
Solve for I_2

$I_2 = I_1 - I_3$

$I_2 = (0.2A) - (-0.1875A)$

$I_2 = 0.387A$

13) An object with a mass of 2×10^{-6} kg and a net charge of q = 500 μC makes a circular path in a magnetic field, if the magnetic field strength is pointing out of the page and the radius of the circular path is 2.8 m, determine the magnitude of the magnetic field strength that the object is moving in if it has a velocity of 6 m/s and what type of charge does the object have?



$|\vec{a}_c| = \frac{v^2}{r}$

$|\vec{F}_m| = qv|\vec{B}|\sin\theta$ $\quad \theta = 90°$

$\Sigma \vec{F} = \vec{F}_m = m\vec{a}_c$

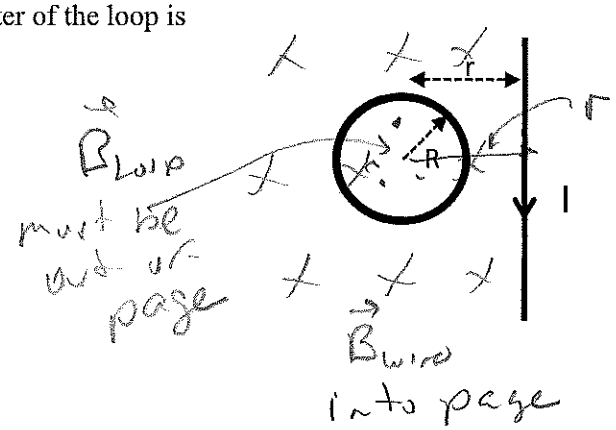
$\Sigma F_r = qv|\vec{B}| = m\left(\frac{v^2}{r}\right)$

$|\vec{B}| = \frac{mv^2}{qvr} = \frac{mv}{qr}$

$|\vec{B}| = 0.0086\,T$

positive – RHR #1

14) A single current-carrying circular loop of radius R = 0.2 m is placed next to a long, straight wire r = 0.4 m from the center of the loop, as shown in the figure below. The current in the wire flows downward towards the bottom of the page and has a magnitude I = 30 A. What must the magnitude and direction (cw or ccw) of the current in the circular loop be so that the net magnetic field at its center of the loop is zero?



$\Sigma \vec{B} = \vec{B}_{wire} + \vec{B}_{loop} = 0$

$\Sigma B_z = |\vec{B}_{wire}| - |\vec{B}_{loop}| = 0$

$\frac{\mu_0 I_w}{2\pi r} - \frac{\mu_0 I_{loop}}{2R} = 0$

From wire to cent of loop

Solve For I_{loop}

$I_{loop} = \frac{2R\mu_0 I_w}{2\pi r \mu_0} = \frac{I_w R}{\pi r}$

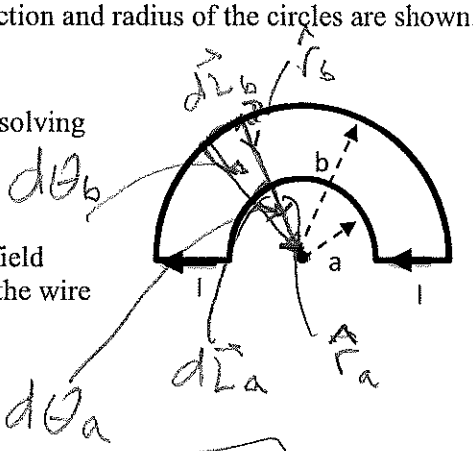
$= 4.77\,A$

ccw – RHR #2

15) A semi-circular current (I) carrying loop is shown below, the current direction and radius of the circles are shown; the smaller loop has a radius of "a" and the larger has a radius of "b".

A) Draw the $d\vec{L}$ and radius ($\vec{r}$ and $\hat{r}$) for each on the diagram along with solving for the dL vector.

B) Derive the equation for the magnitude and direction of the magnetic field strength a point P (center of semi-loops). Ignore the flat portions of the wire in your magnetic field calculations.



$$\vec{B} = \frac{\mu_0 I}{4\pi}\int \frac{d\vec{L}\times\hat{r}}{r^2}$$

$$\Sigma\vec{B}_P = \vec{B}_a + \vec{B}_b$$

$d\theta_a = \frac{dL}{a} \rightarrow dL = a\,d\theta_a$ $\qquad r_a = r = a$

$d\vec{L}\times\hat{r} = |d\vec{L}||\hat{r}|\sin\phi$ $\qquad \phi = 90°$

RHR #1

$d\vec{L}\times\hat{r} \rightarrow +\hat{z}$ direction out of page

$d\vec{L}\times\hat{r} = (dL)\hat{z}$

$$B_{za} = \frac{\mu_0 I}{4\pi}\int_0^{\pi}\frac{a\,d\theta_a}{a^2} = \frac{\mu_0 I}{4\pi a}\int_0^{\pi} d\theta_a = +\frac{\mu_0 I}{4a}$$

$d\theta_b = \frac{dL}{b}$ $\qquad dL = b\,d\theta_b$ $\qquad r_b = b$

$d\vec{L}_b \times \hat{r}_b = |dL_b = b\,d\theta_b$

into page $\rightarrow$ RHR #1

$$|\vec{B}_b| = \frac{\mu_0 I}{4\pi}\int\frac{d\vec{L}_b\times\hat{r}}{r^2}$$

$$B_{zb} = \frac{\mu_0 I}{4\pi}\int_{\pi}^{0}\frac{b\,d\theta_b}{b^2} = \frac{\mu_0 I}{4\pi b}\int_{\pi}^{0} d\theta_b = -\frac{\mu_0 I}{4b}$$

into page

$$\Sigma\vec{B} = \left[\frac{\mu_0 I}{4a} - \frac{\mu_0 I}{4b}\right]\hat{z} = \left[\frac{\mu_0 I}{4}\left(\frac{1}{a}-\frac{1}{b}\right)\right]\hat{z}$$